Unit 10 - Communication Technologies
Assignment 5 - Upgrade a LAN Tasks 3&4

Introduction

Task description - *You are a junior consultant for T5 Alliance. Your new client, Competition Bikes Ltd, have decided that they would like to have wireless capability in their new network. You must adjust your design to include the following new requirements;*

- *Wireless coverage must extend to the whole site.*
- *The wireless network must be as secure as possible.*

Task 1

(U10M1) - Explain why communication protocols are important

Task description - *Explanation of why it is necessary to have networking protocols in a computer network.*

Definition of protocols

Network protocols are sets of rules for exchanging information between hardware devices found within a computer network. This exchange usually occurs much like a dialog between two computers, commonly seen as a type of language between computers.

Essentially, a network protocol is a network communication standard between the hardware found on a network, which are usually established by the organisation IEEE and other similar organisations.

To elaborate, there are many types of protocols and most of them are related to each other because other organisations always produce new networks according to the new technologies, also they use old protocols to make it work for the new devices but now a day's no one uses the old protocol because it's out of fashion.

The network traffic often begins with the client sending a signal to the server, providing key information about what kind of data is being requested. Which means that devices participating in network communication should know about the common network protocol and communicate according to the way, defined by the network protocol. In other words, standard network protocol software must be running on both devices participating in network communication.

For instance, the standard protocols commonly found in a network are UDP, ARP, TCP, IP, FTP and SMTP. TCP/IP is the most simple and common protocols which is actually connected together. SMTP and FTP are based on the TCP/IP model layer.

SMTP Is the easiest and fastest type of protocol to send an email. For example, if the email meets a break through the process it will keep trying to send the email across the network. Therefore, the user wouldn't have to bother about having to resend the email.

Overall, a protocol is basically a set of rules for devices to communicate with each other on a network. Different networks are used for different system to communicate and talk to each other. Protocols are used to communicate and to reduce the error for recovery.

Why protocols are needed

Different network protocols are necessary due to the importance of communication on a network, giving the hardware found in a network the ability to communicate and understand each other.

Essentially, networks act like different languages and they are used to communicate through network standards. Meaning that the network will not work properly because they cannot communicate until the device uses the same language as network, making it very difficult to use the network and sometimes overall impossible to use it.

One of the network organisations, called the Institute of Electrical and Electronics Engineering (IEEE), have rules which orders the different devices on how to communicate. If these rules aren't followed, the user will most likely be disconnected by the server, or vice versa, depending on what the protocol specifications state.

With these regulations by organisations, such as the IEEE, means that it is important to use the correct protocols due to the fact that some protocols are more compatible than others. For instance, if there is one network standard used by user, then there will be no way out for the packet to go to other networks.

So, if people use the correct protocol for the network, it could bring improvement to the performance of the data being trafficked within the network. Which is important when users are communicating over the network, because they have to make sure that the devices they are using have the same language, otherwise the device will not understand the language.

However, there are protocols which can carry out the same function, but do it in a different method or just aren't compatible with other protocols. For example, some exchanges between computers are in short bursts, such as HTTP. While others can stream for long periods of time, usually found in instant messaging.

Basically, the server may send a bundle of data and then close the connection, or it may continue to interact with the client computer until the client decides to end the conversation. It is

this variety of protocols that can be used in specific networks, fitting the intended purpose of certain networks. Especially when connected to a larger network such as the World Wide Web.

Overall, protocols are set of rules which most of the devices uses to send the packet to each other on the different direction without losing it, with organisations such as the IEEE defining protocols so that it could be used as the standard when OEMs are manufacturing the product.

This is so that it is able to talk to other devices made by other manufacturers. Meaning that it keeps the data transfer manageable and certain protocols help computers identify themselves on the Internet.

Task 2

(U10M3) - Assess the effectiveness of data transfer over wired and wireless networks

Task description - *Comparison of wired and wireless networks in terms of;*

- *Performance\speed of transmission*
- *Mobility*
- *Security*
- *Cost to implement*

	Wired	Wireless
Performance	Wired LAN offers better performance as compared to wireless networks. Due to wired connection being consistent up until it's limited range, which is usually 100m. Which would then gradually reduce in performance, meaning that the workstation would have to be within its recommended range in order to fully utilise the speed. Typically, wired network can offer 100Mbps bandwidth using Fast Ethernet technology, and with 1000Mbps using Gigabit Ethernet technology.	Wireless LAN offers slightly worse performance when compared to wired networks. Which is due to the fact that radio frequencies weaken with distance from the wireless access point, especially higher frequencies. Meaning, the further away the user is from the wireless access point, the less performance the user will get from the wireless network. However, the bandwidth in wireless network can vary depending on the frequency of the radio waves.

	Overall, wired networks were typically seen as faster than wireless networks, but wireless internet has been matching the speeds on which wired networks can carry out. However, there are some wired networks which are still faster than the wireless, such as fibre optic reaching speeds of 1Gbps. Also, the requirements of a physical wire has a limited range when compared to wired. Which would cause it lose its performance after that range limit.	For instance, wireless networks with a 2.4 Ghz usually offer speeds of 600 Mbps. While wireless networks with a wireless access point with 5Ghz offering around 1300Mbps. Overall, wireless networks used to be slower than wired LANs, but over the years, wireless technology has improved. Especially with 2.4GHz and 5GHz radio frequency wireless access points improving. However, the range on the faster 5Ghz frequency has a much smaller range than the slower 2.4Ghz.
<u>Mobility</u>	Wired LAN has very poor mobility when compared to wireless. The reason the mobility is very poor is due to the fact that wired networks require physical cables to connect every workstation on the network. Also making those devices without an ethernet port obsolete in a wired network. Therefore meaning that running Ethernet cables between rooms or floors can be a significant challenge to those setting up and maintaining a network. However, having a lack of mobility gives the admins a good idea of where the devices are at all times. Meaning that fixing a hardware issue with a specific device in a network would be a lot easier to find for the admins.	Wireless LAN has very good mobility when compared to wireless. The reason the mobility is very good is due to the fact that a wireless network has one or more access points which multiple devices could connect to, without requiring a ethernet port. Instead requiring an antenna, which is usually factory standard with modern devices. Therefore meaning that mobile computing devices, such as smartphones and laptops, can be moved around freely within a wireless network. However, this does not mean the mobility on wireless network covers the entire building, due to its limited range.

	Overall, wired networks don't have good mobility, due to the network being made of cables. Since all workstations on a wired network must be connected by Ethernet cable. It makes the workstations stationary in order to access the network, removing the concept of mobility overall.	Meaning that multiple wireless access points are needed for each room, or users are forced to stay in the range of the limited amount of wireless access points. Which brings multiple problems to the wireless network, such as performance loss or the wireless access point malfunctioning due to the overload of data. Overall, wireless network have very good mobility, due to there being one or more access points to connect to. Meaning that multiple devices with an antenna installed can connect to the network. Making mobility much more flexible than wired, since the user doesn't have to stay stationary to access the network.
Security	Wired LAN has very good security when compared to a wireless network. A wireless LAN can also have a security system which can be considered just as good, but only when the wired network is connected to the internet. The reason the security is good for wired networks is due to the fact that they are secluded from outside access, making it impossible to access them unless the user physically connects a wire. However, it is possible for a wired network to be connected to the internet, and the security considerations for a wired network	Wireless LAN has good security, but wired LAN also has a very good security system. The reason that security is good for wireless networks is that it uses wired equivalent privacy, or WEP. Which is the encryption of passwords to access the wireless network, to protect the data. This is what makes wireless networks as secure as wired networks if connected to the internet. However, the security of wireless network isn't as good as a wired network without connection to the internet.

	connected to the internet are firewalls. Firewall software can be installed on each computer. Therefore meaning that running Ethernet cables between rooms or floors can be a significant challenge to those setting up and maintaining a network. However, having a lack of mobility gives the admins a good idea of where the devices are at all times. Meaning that fixing a hardware issue with a specific device in a network would be a lot easier to find for the admins. Overall, wired networks have good security, due to the network being secluded, and having a firewall when connected to the internet. However, the firewall doesn't completely block out access to the network, since it isn't possible. Which is what makes it equally as good as the WEP on the wireless network. On the other hand, if the wired network isn't connected to the internet, it has the best security possible for a network due to the isolation of the network from outside access.	Due to the fact that the security of a wireless network can still be intercepted. While a secluded wired network is impossible to intercept unless physically connected. Overall, wireless network security is good due to the WEP encryption system, significantly increasing the security of data within the network. However, it doesn't make it completely secure from outside access like the isolated wired network that isn't connected to the internet. Meaning that the network still has a risk of being intercepted by an outside source. Making wired networks much better at security when compared to wireless, but only if the wired network isn't connected to an outside network, such as the World Wide Web.
<u>Cost</u>	The cost of a wired LAN is less when compared to wireless network when dealing with a small network, but a large network could cost more than wireless.	The cost of a wireless LAN is a lot more when compared to a wired network, but it could cost less than a wired network when dealing with a large network.

	The reason the security is good for wired networks is due to the fact that Ethernet cables and switches are not expensive when compared to the recommended hardware of a wireless network, such as wireless access points. However, the price of all the ethernet cables and switches can add up the larger the network gets, making the budget increase over the budget of a wireless network with similar size. Therefore meaning that purchasing more and more cables and switches can be a challenging to those setting up and maintaining a network, due to the amount of cables and hardware. Which also makes it very prone to damaging the cables or the hardware, adding more to the budget due to the repair or replacement costs. Overall, wired networks are typically seen as cheaper than wireless networks, especially on smaller networks. However, wired networks of larger size can be less profitable than wireless networks of a large size. Also, the costs of the repairs and damage of the cables and switches, which become much more likely to happen due to the increasing number of obstacles. Which would cause the budget to increase significantly over time, especially when compared to wireless networks.	The reason that costs are more expensive for a smaller network is due to the fact that wireless connection points can cost more than ethernet cables and switches. However, on larger networks, not many more wireless access points are needed to cover the range needed for the devices and workstations within the building. Making the budget increase much slower when compared to a wired network, due to multiple users being able to access a single access point without cables. Therefore meaning that purchasing more wireless access points to satisfy the recommendations becomes less expensive compared to that of a wired network. However, a lot of skill and knowledge is required to set up a wireless network, due to the complexity of a wireless network. Which means that staff that are highly trained in networking are needed to set them up, and since they are highly trained, they require a higher salary from the company. Overall, wireless networks are typically seen as more expensive than wireless networks, when dealing with smaller networks. However, wireless networks of larger size can be more profitable than wired networks of a large size. However, the costs of getting the staff to set up a wireless network could cost more than a wired network, due

		to the skills and knowledge needed to setup the wireless network. Which would give a large budget initially, but over time the budget would be more consistent without many peaks in cost. Unless one of the wireless access points malfunctions and needs a repair or replacement, which is much more rare than a malfunction of a wired network.

Conclusion

In conclusion, both wireless and wired networks have their own advantages and disadvantages. Meaning that, to benefit out of either wireless or wired network, would depend on the type of network that needs to be built.

If the company is dealing with a small network, it would be better to invest on a wired network, due to the advantages a wired network gives. The reason for this is due to the fact that small wired networks are less expensive, faster, and more secure than small wireless networks. With small wireless networks are more expensive, slower over distance, and, in some cases, less secure than wireless networks.

However, if the company was to invest in a larger network, it would be much better to build a wireless network, due to the advantages a wireless network gives. The reason for this is due to the wireless network being less expensive to set up and having a more predictable budget over time, better reliability with a large group of workstations and the ability for users to access the network with other devices, instead of always investing in workstations.

With larger wired networks costing more than a wireless network over time (sometimes initially), less reliable due to the cables being more prone to damage and the complex task of making sure all the workstations are connected to the network through cables, which require a lot of time to check.

Overall, a wired network would be better for a smaller network, due to the security being better, the faster data speeds and costing less than a wireless access point. However, a wireless network would be better for a larger network, due to the costs being less over time and sometimes initially, a larger ratio of users to a single wireless access point to that of a wired network from users to a switch and a wireless network being more reliable than a wired network when dealing with accidental damage and malfunctions.